

MULTIPLEXER AND DEMULTIPLEXER

Aim (i) Implement the following functions using 8:1 mux IC 74151

a) $f_1(A, B, C) = \sum m(2, 4, 6, 7)$

b) $f_2(A, B, C, D) = \sum m(0, 2, 3, 6, 8, 9, 12, 14)$

(ii) Implement 1:4 DEMUX using gates

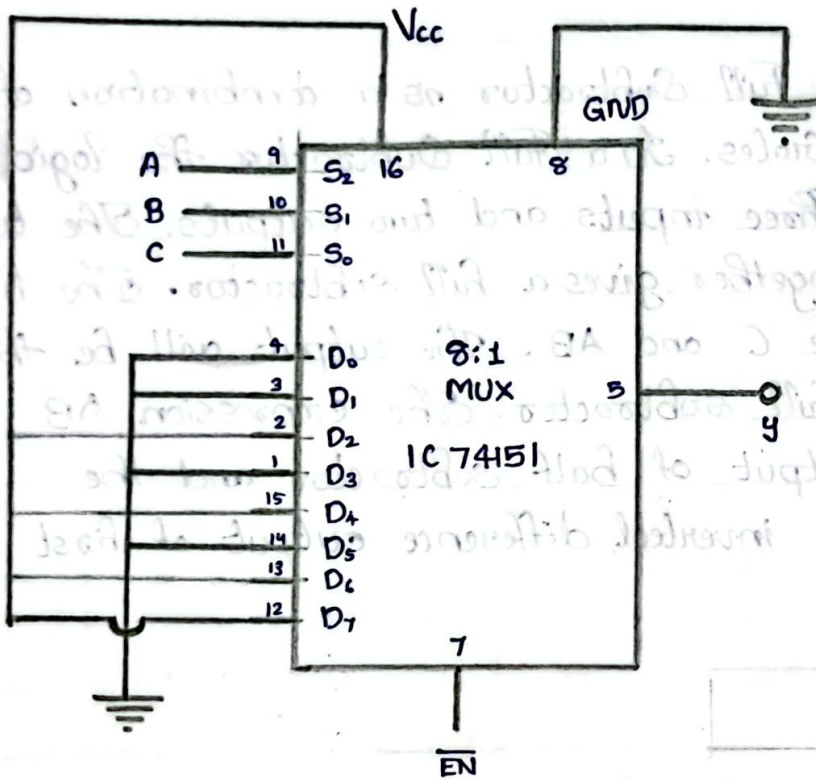
Components Required

S.No.	COMPONENT	SPECIFICATION	QTY
1.	IC 7411	Triple 1P AND GATE	2
2.	IC 7404	NOT GATE	1
3.	IC 74151		1
4.	IC trainer kit		1
5.	IC tester		1

Theory

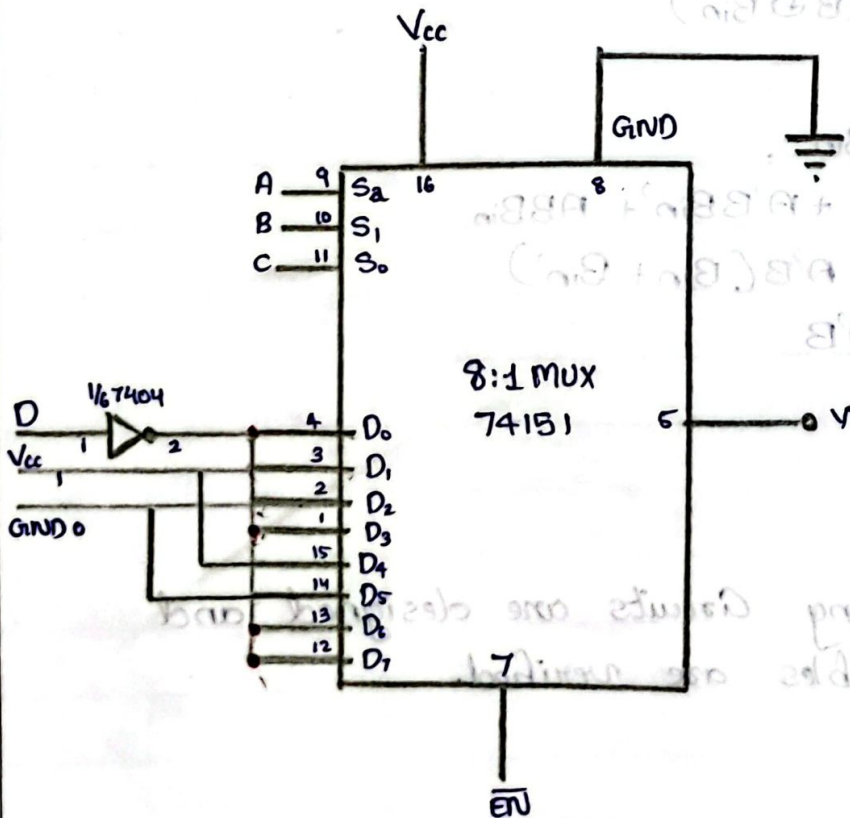
Multiplexers are logical circuits which select the only one digital information from several inputs. It is also known as data selector as it selects only one output from many input. Multiplexing is a process of translating large number of information through a single line if there are m select lines, there will be 2^m data lines.

a) $f(A, B, C) = \sum m(2, 4, 6, 7)$



A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

(b) $f(A, B, C, D) = \sum m(0, 2, 3, 6, 8, 9, 12, 14)$



A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

Demultiplexers are the logic circuits which provides several informations from a single data input.

Demultiplexing is the process of taking information from one input and transmitting the same over several outputs. If there is n input lines then 2^n output lines will be there. The select line determine to which output line the data input should be given.

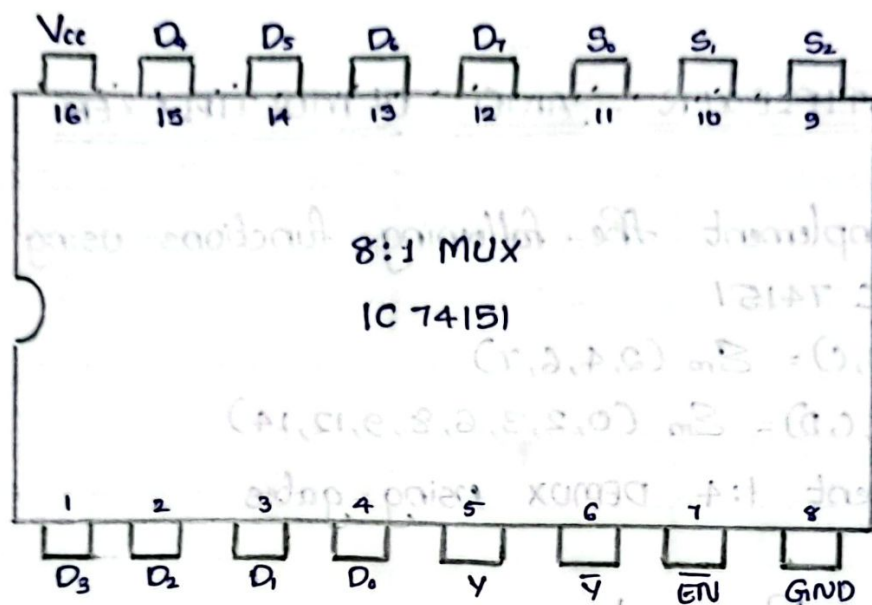
Procedure :-

- Check whether all the ICs are working properly.
- Connect the circuit as shown in figure.
- Drive the supply voltage and ground.
- Obtain the output and verify the output using truth table.

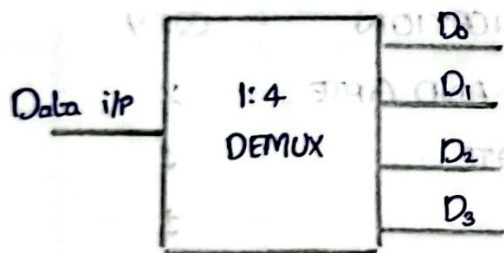
Result :-

Realisation of multiplexer and demultiplexer are done and verified the truth table.

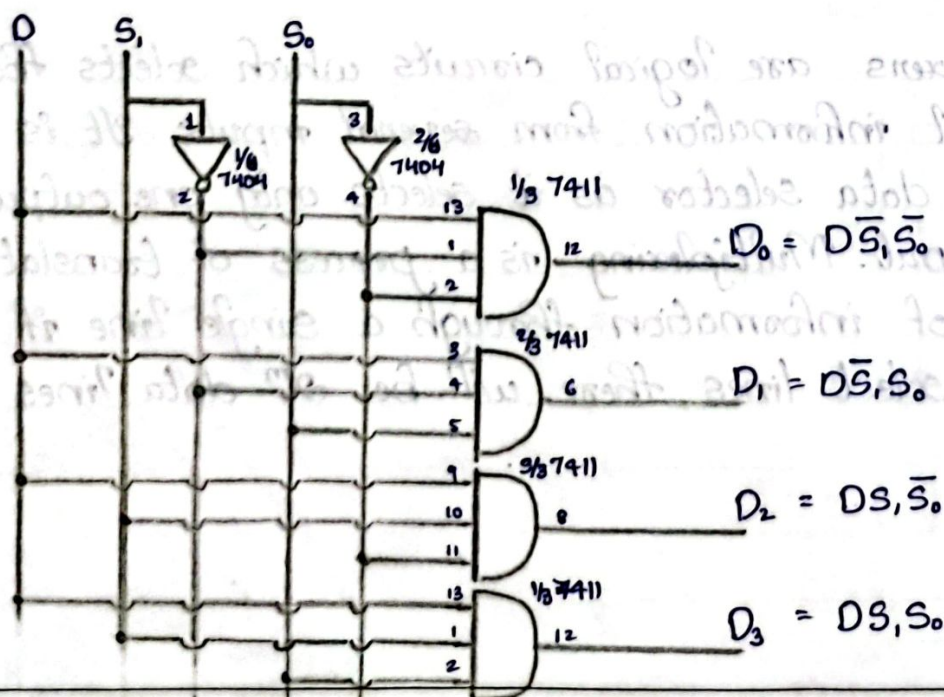
PIN DIAGRAM



(ii) 1:4 DEMUX



Select i/p		O/P			
S ₁	S ₀	D ₀	D ₁	D ₂	D ₃
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D



CODE CONVERTERS

Aim - To design and setup the following circuits.

- ① A 4-bit Binary to Gray code converter
- ② A 4-bit Gray to Binary code converter
- ③ A 3-bit Binary to Gray and Gray to Binary Code converter using mode control.

Components :

Sl.No	Components	Specification	Quantity
1	IC 7486	X-OR Gate	3
2	IC 7408	AND Gate	1
3	IC Trainer kit		

Theory -

To convert a Binary number to corresponding Gray code, the following rules are applied.

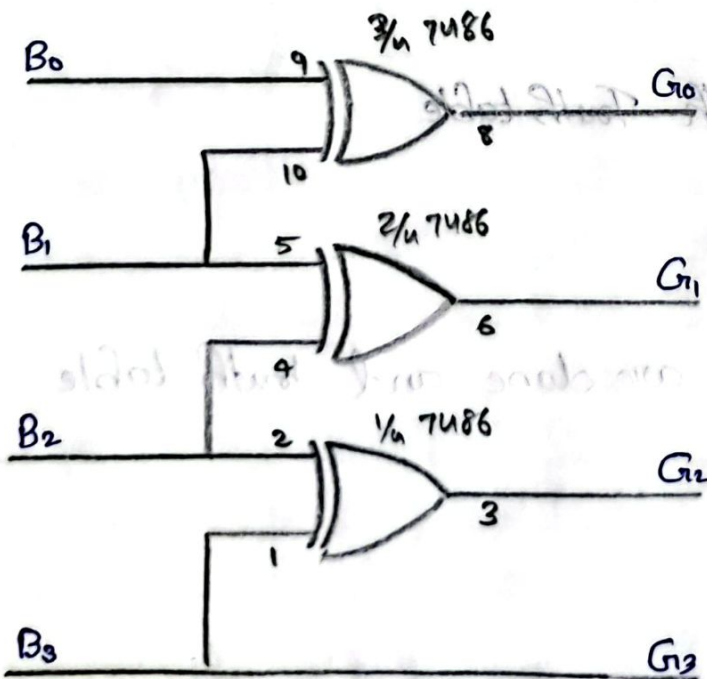
1. The MSB in the Gray Code is the same as the corresponding bit in a Binary number.
2. Going from left to right, add each adjacent pair of binary digits to get the next Gray code digit. Disregard carries.

As the first step to design a Binary to Gray code converter, setup a truth table with binary number $B_3 B_2 B_1 B_0$ and corresponding Gray code numbers $G_3 G_2 G_1 G_0$. Setup a circuit

Binary to Gray

Binary				Gray			
B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	0	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	0
1	0	1	1	1	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	0
1	1	1	1	1	0	1	1

e/p



$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$G_3 = B_3$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

$$G_2 = B_3 \oplus B_2$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	1	0	0
10	0	0	1	1

$$G_1 = B_2 \oplus B_1$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

$$G_0 = B_1 \oplus B_0$$

realising the simplified logic expression obtained using K-Maps for G_i s as the functions of B_i s.

To convert from Gray to Binary, the following rules are applied.

1. The most significant digit in the binary number is the same as the corresponding digit in the Gray code.
2. Add each binary digit generated to the Gray code digit in the next adjacent position. Disregard carries.

To design the Gray to Binary code converters, setup the truth table and get simplified expressions using K-Maps for each binary bits as a function of Gray code bits. Each Gray code number differs from the preceding number by a single bit.

A Gray to Binary and Binary to Gray code converter is setup using a mode control pin M such that when $M=0$ it functions as Binary to Gray code converter and when $M=1$ it functions as a Gray code to Binary converter.

Gray to Binary

Gray				Binary			
G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	0	0	0
1	0	1	1	1	0	0	1
1	1	0	0	1	0	1	1
1	1	0	1	1	0	1	0
1	1	1	0	1	1	0	1
1	1	1	1	1	1	0	0

$G_3 G_2$ \ $G_1 G_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$B_3 = G_3$$

$G_3 G_2$ \ $G_1 G_0$	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

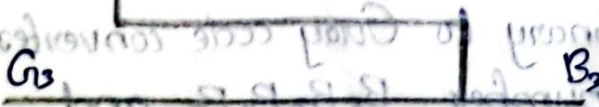
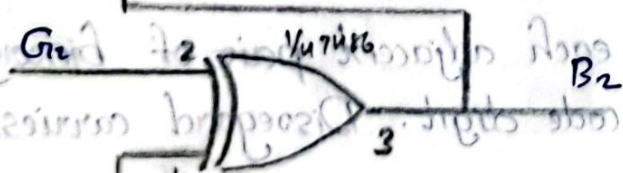
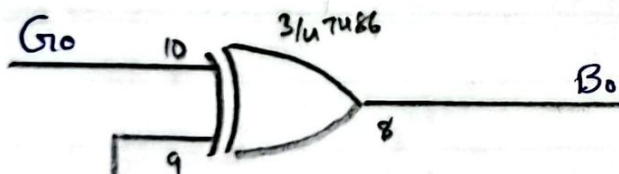
$$B_2 = G_3 \oplus G_2$$

$G_3 G_2$ \ $G_1 G_0$	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

$$B_1 = G_3 \oplus G_2 \oplus G_1$$

$G_3 G_2$ \ $G_1 G_0$	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

$$B_0 = G_3 \oplus G_2 \oplus G_1 \oplus G_0$$



e/p

Procedure:

1. Test all the components and IC packages using multimeter and digital IC tester.
2. Verify the truth tables of the circuit by feeding the input bit combinations.

Result &

Binary to Gray code converter and Gray to Binary code converter is designed and truth table verified.

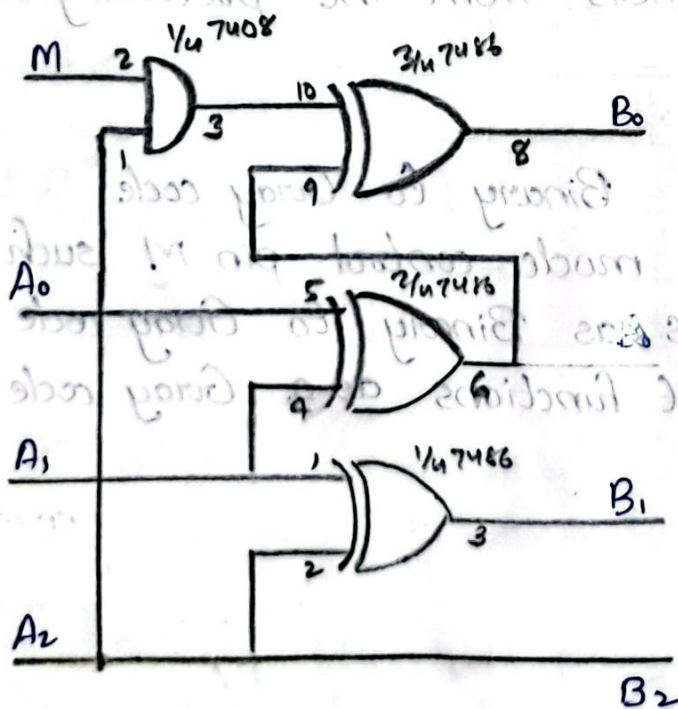
B ₃ B ₂		B ₁ B ₀		G ₃ G ₂		G ₁ G ₀	
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	1	0	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	0
1	0	1	1	1	1	1	1
1	1	0	0	0	0	0	0
1	1	0	1	0	0	0	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	1	1

M	A ₂	A ₁	A ₀	B ₂	B ₁	B ₀
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	1	0	0
0	1	0	1	1	0	1
0	1	1	0	1	1	0
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	0	1	1
1	1	0	0	1	0	0
1	1	0	1	1	0	1
1	1	1	0	1	1	0
1	1	1	1	1	1	1

A ₁ A ₀ \ MA ₂	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$B_2 = A_2$$

A ₁ A ₀ \ MA ₂	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	1	0	0
10	0	0	1	1



A ₁ A ₀ \ MA ₂	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	1	0	1	0
10	0	1	0	1

$$\begin{aligned}
 B_0 &= \bar{M} \bar{A}_1 \bar{A}_0 + \bar{M} A_1 \bar{A}_0 + \\
 &\quad \bar{A}_2 \bar{A}_1 A_0 + \bar{A}_2 A_1 \bar{A}_0 \\
 &\quad + M A_2 \bar{A}_1 \bar{A}_0 + M A_2 A_1 A_0 \\
 &= (M A_2) \oplus A_1 \oplus A_0
 \end{aligned}$$